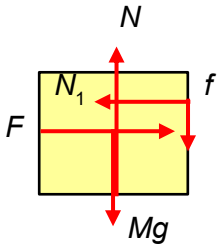
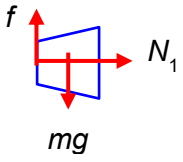


2	
(a)	Newton's 2 nd law states that the <u>rate of change of momentum</u> is <u>directly proportional</u> to the <u>net external force</u> and is <u>in the direction of this force</u> .
(b)(i)1	
	Label type of forces N – normal contact force on block by surface N_1 – normal contact force on block by beaker f – frictional force on block by beaker Mg – gravitational force/weight on block by earth
	4 correct points of application Correct length of arrows • $F > N_1$ • $N = Mg + f$
(b)(i)2	
	Label type of forces N_1 – normal contact force on beaker by block f – frictional force on beaker by block mg – gravitational force/weight on beaker by earth
	3 correct points of application (line of action of N_1 is below the CG)

	Correct length of arrows • $f = mg$ • N_1 on block = N_1 on beaker • f on block = f on beaker
(b)(ii)	Given $f = \mu N_1$ $\quad = 0.40 N_1 \quad \dots\dots(1)$ By Newton's 2nd law, $ma = N_1 \quad \dots\dots(2)$
	As in vertical equilibrium, $f = mg \quad \dots\dots(3)$
	$0.40 N_1 = mg$ $0.40 ma = mg$ $a = \frac{g}{0.40} = \frac{9.81}{0.40}$ $\quad = 24.53 \text{ m s}^{-2} \text{ (4 sf)}$ $\quad = 24.5 \text{ m s}^{-2} \text{ (3 sf)}$
(b)(iii)1	For beaker to be in equilibrium (in position) normal to the slope, $f_2 = mg \cos 50^\circ$ $\mu N_2 = mg \cos 50^\circ$ $(0.40) N_2 = (0.10)(9.81) \cos 50^\circ$
	$N_2 = 1.58 \text{ N (3sf)}$
(b)(iii)2	<u>Actual acceleration</u> down the slope due to weight, $a_p = g \sin 50^\circ$ $\quad = 7.51 \text{ m s}^{-2}$
	Acceleration along the slope for beaker to be in position <u>required</u>, $ma_2 = mg \sin 50^\circ + N_2$ $(0.10) a_2 = (0.10)(9.81) \sin 50^\circ + 1.58$ $a_2 = 23.3 \text{ m s}^{-2} \text{ (3sf)}$
	Since $a_p < a_2$, the beaker cannot remain in the position.
	OR
	To remain in position, there must be frictional force $f_2 = mg \cos 50^\circ$
	The block and beaker are both accelerating along the slope due to their weight $a_p = g \sin 50^\circ$ $\quad = 7.51 \text{ m s}^{-2}$
	As both have the same acceleration, their normal contact force N_2 must be 0, and hence frictional force f_2 is also 0.
	Hence, the beaker cannot remain in the position.

Anglo-Chinese Junior College
H2 (9749) Physics

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